

Software Development Life Cycle (SDLC) & Agile Methodology

Detailed Trainer Notes

SDLC Fundamentals

Software Development Life Cycle (SDLC) is a structured process used to design, develop, test, deploy and maintain software. It improves quality, reduces risk, and ensures predictable delivery. Typical phases include Planning, Requirements, Design, Development, Testing, Deployment and Maintenance. Each phase has specific deliverables, stakeholders, entry/exit criteria and quality gates.

SDLC Models

Common models include Waterfall, V-Model, Iterative, Incremental, Spiral and Agile. Waterfall suits stable requirements, while Agile supports changing requirements. Spiral emphasizes risk analysis and Iterative delivers software in repeated cycles.

Agile & Scrum

Agile values customer collaboration, responding to change, working software and individuals/interactions. Scrum roles include Product Owner, Scrum Master and Developers. Scrum events are Sprint Planning, Daily Scrum, Sprint Review and Sprint Retrospective. Artifacts include Product Backlog, Sprint Backlog and Increment.

DevOps & CI/CD

DevOps combines development and operations using automation. CI frequently integrates code changes with automated builds and tests. CD automates deployments to staging or production. Common tools include Git, Jenkins, Docker, Kubernetes, SonarQube and Maven.

Case Study

A banking application team receives changing compliance requirements during development. Explain why Agile is preferred over Waterfall, identify Scrum roles, define backlog items, design a CI/CD pipeline, and explain testing activities during each sprint.